Hop In Express

COMMAND OS

Launch Package
February 16, 2026

Command OS: Administrator Operations Manual

System Configuration, Monitoring, and Security ---

🔐 Security Configuration

Role-Based Access Control (RBAC)

Command OS uses granular role definitions in `lib/rbac.ts`. Ensure users are assigned correct roles:

- **Owner**: Full access. Can modify system settings.
- **Manager**: Can edit Rota, Approve Leave, View Audit Logs.
- **Cashier**: Sales only. Cannot view financial reports or modify stock.
- **Assistant**: Restricted access. View tasks only.

Biometric Enrollment (Face Auth)

1. Navigate to **Staff > Profile > Security**. 2. Tap "Enroll Face". 3. Ensure good lighting. Capture front, left, right angles. 4. Test unlock immediately. ---

💾 Data Management (Firestore)

Backup Strategy

- **Automated**: Firestore handles daily snapshots (Standard Google Cloud Backup).
- **Wait**: Do not solely rely on snapshots.
- **Manual Export**:

1. Go to **Project Settings > Integrations**. 2. Select "Export Data" (Download JSON). 3. Store securely offline (e.g., Encrypted USB).

Reporting

- **BigQuery Export**: (Planned Phase 2) Auto-sync for heavy analytics.
- **Performance Monitoring**: Check Firebase Console > Performance tab weekly.

🛠️ Troubleshooting & Support

Common Issues

- **"Permission Denied"**: Check user role in `StaffView`. Ensure strict equality (e.g., 'Manager' vs 'manager').
- **Sync Conflicts**: If multiple devices edit the same record offline, "Last Write Wins". Train staff to sync frequently.

- **Camera Failure**: Ensure browser has camera permissions allowed (HTTPS context required).

🚀 Future Roadmap (ML Integration)

See `docs/ML_ROADMAP.md` for detailed plans on: 1. Predictive Inventory Replenishment. 2. Smart Rota Optimization. 3. Computer Vision Shelf Audits.

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